

AY: 2025-2026

EXAM | NLP

June 2026

M1-S2: Dept. of Electrical Engineering

Teacher: A. Mhamdi

Time Limit: 1½ h

This document contains 7 pages numbered from 1 to 7. Upon receiving it, verify completeness. The 3 tasks are independent and can be solved in any order you prefer. The following rules apply:

- ❶ A handwritten double-sided A4 sheet is permitted.
- ❷ Any electronic material, except basic calculator, is prohibited.
- ❸ Mysterious or unsupported answers will not receive full credit.
- ❹ Round results to the nearest thousandth (i.e., the third digit after the decimal point).
- ❺ Task N°3: Correct answers earn points as indicated. There is no negative scoring.

### Task N°1

⌚ 20mn | (5 points)

Consider the following corpus<sup>1</sup>.

- <s> the cat sat on the mat </s>
- <s> the cat sat </s>
- <s> the dog sat on the log </s>
- <s> the dog barked </s>
- <s> the cat barked </s>

(a) (1½ points) Compute the following probabilities using maximum likelihood estimation:

$P(\text{sat}|\text{the cat}), \quad P(\text{on}|\text{cat sat}), \quad P(\text{barked}|\text{the dog})$

Using MLE:

count(the cat) = 3,    count(the cat sat) = 2     $\Rightarrow P = 2/3$   
 count(cat sat) = 2,    count(cat sat on) = 1     $\Rightarrow P = 1/2$   
 count(the dog) = 2,    count(the dog barked) = 1     $\Rightarrow P = 1/2$

(b) (1 point) Using a trigram model, compute the probability of:

<s> the dog barked on the mat </s>

<sup>1</sup>Term <s> is not included in the token count.

We compute the probability of the considered sentence as follows:

$$\begin{aligned}
 P &= \prod P(w_i | w_{i-2} w_{i-1}) \\
 &= P(\text{the} | \langle s \rangle \langle s \rangle) \times P(\text{dog} | \langle s \rangle \text{the}) \times P(\text{barked} | \text{the dog}) \times P(\text{on} | \text{dog barked}) \\
 &\quad \times P(\text{the} | \text{barked on}) \times P(\text{mat} | \text{on the}) \times P(\langle s \rangle | \text{the mat}) \\
 &= \dots \times \underbrace{P(\text{on} | \text{dog barked})}_{\text{count}(\text{dog barked on})=0} \times \underbrace{P(\text{the} | \text{barked on})}_{\text{count}(\text{barked on the})=0} \times \dots \\
 &= 0
 \end{aligned}$$

**Further analysis:**

→ If a trigram is unseen, we back off to the bigram;

→ if the bigram is unseen, we back off to the unigram.

Using backoff to lower-order n-grams:

$$\begin{aligned}
 \tilde{P}(w_i | w_{i-2} w_{i-1}) &= \begin{cases} P_{\text{tri}} & \text{if seen} \\ P_{\text{bi}} & \text{otherwise} \\ P_{\text{uni}} & \text{if needed} \end{cases} \\
 P &= \prod \tilde{P}(w_i | \cdot) \\
 &= \dots \times \underbrace{P(\text{on})}_{\neq 0} \times \underbrace{P(\text{the} | \text{on})}_{\neq 0} \times \dots \quad (\text{with backoff at missing trigram}) \\
 P(\text{on} | \text{dog barked}) &\rightarrow P(\text{on} | \text{barked}) \rightarrow P(\text{on}) = 2/31 \\
 P(\text{the} | \text{barked on}) &\rightarrow P(\text{the} | \text{on}) = 1
 \end{aligned}$$

(c) (1 point) Using Laplace smoothing, compute:

$$P(\text{mat} | \text{the dog})$$

The vocabulary size is  $|V| = 9$  (set of unique words:  $\{\langle s \rangle, \text{barked}, \text{cat}, \text{dog}, \text{log}, \text{mat}, \text{on}, \text{sat}, \text{the}\}$ ). Using add-one smoothing, we get:

$$P = \frac{0 + 1}{2 + 9} = \frac{1}{11}$$

(d) ( $1\frac{1}{2}$  points) Compare the probabilities of sentences A and B under the trigram model:

A: <s> the cat sat on the mat </s>

B: <s> the dog barked on the mat </s>

$$\text{count}(\text{barked on the}) = 0 \Rightarrow P(B) = 0 \quad \text{and} \quad P(A) > 0$$

Even after smoothing,  $P(B) > 0$  still smaller. The model assigns a higher probability to A than to B.  $P(A) > P(B) \Rightarrow A$  is more likely under the model.

### Task №2

⌚ 40mn | (8 points)

We want to compute the self-attention output for the sequence “cat sat” using the scaled dot-product attention mechanism with positional encodings. For that, suppose that each word is represented by 4-dimensional vector.

cat  $\rightarrow [0.3, -0.7, 1.2, 0.5]$

sat  $\rightarrow [0.9, 0.4, -0.6, 0.8]$

The initial weight matrices are:

$$\mathcal{W}_Q = \mathcal{W}_K = \mathcal{W}_V = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(a) (1 point) Determine the values of  $d_{\text{model}}$  and  $\text{max\_len}$ .

- Embedding dimension:  $d_{\text{model}} = 4$
- Sequence length:  $\text{max\_len} = 2$

(b) (1 point) Compute and add positional encodings to word embeddings.

Positional encoding formula:

$$\text{PE}(\text{pos}, 2i) = \sin\left(\frac{\text{pos}}{10000^{(2i/d_{\text{model}})}}\right)$$

$$\text{PE}(\text{pos}, 2i + 1) = \cos\left(\frac{\text{pos}}{10000^{(2i/d_{\text{model}})}}\right)$$

where:

pos: position in sequence

i: dimension index

**cat** with pos = 0, we have:

$$1^{\text{st}} \text{ dim} \implies \text{PE}(0, 0) = 0$$

$$2^{\text{nd}} \text{ dim} \implies \text{PE}(0, 1) = 1$$

$$3^{\text{rd}} \text{ dim} \implies \text{PE}(0, 2) = 0$$

$$4^{\text{th}} \text{ dim} \implies \text{PE}(0, 3) = 1$$

$\therefore$  Pos 0 encoding is  $[0, 1, 0, 1]$

The new input embedding is:

$$\text{cat: } [0.3, -0.7, 1.2, 0.5] + [0, 1, 0, 1] = [0.3, 0.3, 1.2, 1.5]$$

$$\text{sat: } [0.9, 0.4, -0.6, 0.8] + [0.841, 0.540, 0.01, 1] = [1.741, 0.94, -0.59, 1.8]$$

**sat** with pos = 1, we have:

$$1^{\text{st}} \text{ dim} \implies \text{PE}(1, 0) = 0.841$$

$$2^{\text{nd}} \text{ dim} \implies \text{PE}(1, 1) = 0.540$$

$$3^{\text{rd}} \text{ dim} \implies \text{PE}(1, 2) = 0.01$$

$$4^{\text{th}} \text{ dim} \implies \text{PE}(1, 3) = 1.0$$

$\therefore$  Pos 1 encoding is  $[0.841, 0.540, 0.01, 1]$

(c) (1 point) Compute Q, K and V matrices.

The input matrix  $\mathcal{X}$  is:

$$\mathcal{X} = \begin{bmatrix} 0.3 & 0.3 & 1.2 & 1.5 \\ 1.741 & 0.94 & -0.59 & 1.8 \end{bmatrix}$$

Since  $\mathcal{W}_Q = \mathcal{W}_K = \mathcal{W}_V = \mathbf{I}_4$ :

$$\mathcal{Q} = \mathcal{X} \cdot \mathcal{W}_Q = \mathcal{X}$$

$$\mathcal{K} = \mathcal{X} \cdot \mathcal{W}_K = \mathcal{X}$$

$$\mathcal{V} = \mathcal{X} \cdot \mathcal{W}_V = \mathcal{X}$$

$$\therefore \mathcal{Q} = \mathcal{K} = \mathcal{V} = \begin{bmatrix} 0.3 & 0.3 & 1.2 & 1.5 \\ 1.741 & 0.94 & -0.59 & 1.8 \end{bmatrix}$$

(d) (2 points) Compute attention scores.

The formula to compute the attention scores is:

$$\text{Attention Score} = \frac{\mathcal{Q} \cdot \mathcal{K}^T}{\sqrt{d_{\text{model}}}}$$

The attention matrix (*before softmax*) is:

$$\text{Attention}_{\text{before}} = \begin{bmatrix} \underbrace{1.935}_{\text{score(cat, cat)}} & \underbrace{1.398}_{\text{score(cat, sat)}} \\ \underbrace{1.398}_{\text{score(sat, cat)}} & \underbrace{3.751}_{\text{score(sat, sat)}} \end{bmatrix}$$

The attention matrix (*after softmax*) is:

$$\text{Attention}_{\text{after}} = \begin{bmatrix} 0.631 & 0.369 \\ 0.087 & 0.913 \end{bmatrix}$$

(e) (2 points) Interpret the findings.

$$\text{Attention}_{\text{after}} = \begin{bmatrix} \underbrace{0.631}_{\text{cat attends 63.1\% to itself}} & \underbrace{0.369}_{\text{cat attends 36.9\% to sat}} \\ \underbrace{0.087}_{\text{sat attends 8.7\% to cat}} & \underbrace{0.913}_{\text{sat attends 91.3\% to itself}} \end{bmatrix}$$

(f) (1 point) Apply attention to values.

The final output is  $\text{Attention} \cdot V$

$$\text{Output} = \begin{bmatrix} 0.832 & 0.536 & 0.539 & 1.611 \\ 1.616 & 0.884 & -0.434 & 1.774 \end{bmatrix}$$

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Teacher: A. Mhamdi

Full Name: .....

ID: .....

Class: RAIA1 .....

Room: .....

Time Limit: 1½ h

ANSWER SHEET

Task N°3

⌚ 30mn | (7 points)

- (a) (½ point) What is the primary architecture that most modern LLMs are based on?
- ☐ CNN ☐ RNN ☐ LSTM ☒ Transformers
- (b) (½ point) What is “hallucination” in LLMs?
- ☒ When the model generates factually incorrect or nonsensical information
- ☐ The model’s ability to generate creative content
- ☐ The model’s visualization capabilities
- ☐ When the model processes images along with text
- (c) (½ point) Which Python library is used for deep learning in NLP tasks?
- ☐ Scikit-learn ☒ PyTorch ☐ Matplotlib ☐ Beautiful Soup
- (d) (½ point) Which of the following is a common method for word embedding in NLP?
- ☐ One-hot encoding ☐ Bag of words ☐ TF-IDF ☒ Word2Vec
- (e) (½ point) What does “\d+(?= dollars)” match?
- ☐ A number followed by “ dollars” (including “ dollars”)
- ☐ A number not followed by “ dollars”
- ☒ Only the numbers in “100 dollars” but not “100 euros”
- ☐ It is invalid regex syntax
- (f) (½ point) What is the value of the expression 5 + True - False \* None?
- ☐ 6 ☐ 5 ☒ Error ☐ None
- (g) (½ point) What is a “language model” in NLP?
- ☒ A probability distribution over sequences of words
- ☐ A tokenization algorithm
- ☐ A text summarization tool
- ☐ A set of predefined rules

✂

- (h) ( $\frac{1}{2}$  point) What is a Recurrent Neural Network (RNN)?
- ☐ A type of neural network that can only handle static inputs
  - ☐ A type of neural network that is used for image recognition
  - ☒ A type of neural network that processes sequences of inputs
  - ☐ A type of neural network that is used for unsupervised learning
- (i) ( $\frac{1}{2}$  point) Which of the following is a limitation of RNNs?
- ☐ They can handle long-term dependencies easily
  - ☐ They can be trained in parallel
  - ☒ They suffer from the vanishing gradient problem
  - ☐ They are highly scalable
- (j) ( $\frac{1}{2}$  point) What is the primary purpose of the LSTM (Long Short-Term Memory) cell?
- ☒ To address the vanishing gradient problem in RNNs
  - ☐ To increase the speed of training RNNs
  - ☐ To reduce the number of layers in a neural network
  - ☐ To make the network more complex
- (k) ( $\frac{1}{2}$  point) What does the self-attention mechanism in Transformers allow the model to do?
- ☒ Consider the importance of each word in a sequence when processing another word
  - ☐ Ignore the context of words in a sequence
  - ☐ Focus only on the last word in a sequence
  - ☐ Process sequences in a sequential manner
- (l) ( $\frac{1}{2}$  point) If you needed to perform topic modeling on a corpus of documents, which library would be most suitable?
- ☐ NLTK
  - ☐ Polyglot
  - ☐ spaCy
  - ☒ Gensim
- (m) ( $\frac{1}{2}$  point) Which keyword is used for creating an anonymous function in Python?
- ☒ lambda
  - ☐ def
  - ☐ fun
  - ☐ anonymous
- (n) ( $\frac{1}{2}$  point) In Python, what does the \*args syntax in function definition mean?
- ☒ It allows passing a variable number of arguments
  - ☐ It defines a default argument
  - ☐ It indicates keyword arguments
  - ☐ It allows only integer arguments